

HTML Basics Notes

1. What is HTML?

HTML stands for HyperText Markup Language.

It is the standard markup language used to create web pages.

HTML describes the structure of a webpage using elements (tags).

2. Basic HTML Structure

```
<!DOCTYPE html>
<html>
  <head>
    <title>My First Web Page</title>
  </head>
  <body>
    <h1>Hello, World!</h1>
    <p>This is a paragraph.</p>
  </body>
</html>
```

Explanation:

<!DOCTYPE html>: Declares HTML5 document.

<html>: Root element.

<head>: Metadata (e.g., title, styles).

<body>: Visible content.

3. Common HTML Tags

Text Formatting Tags

<h1> to <h6>: Headings (h1 is largest).

<p>: Paragraph.

: Bold text.

: Italic text.

: Line break.

Lists

Unordered List (bullets)

```
<ul>
  <li>Item 1</li>
  <li>Item 2</li>
</ul>
```

Ordered List (numbers)

```
<ol>
  <li>First</li>
  <li>Second</li>
</ol>
```

4. Links and Images

Anchor Tag

```
<a href="https://example.com">Visit Example</a>
```

href attribute defines the link URL.

Image Tag

```

```

src: Image URL/path.

alt: Alternate text.

width/height: Dimensions.

5. Forms in HTML

```
<form action="/submit" method="post">
  <label for="name">Name:</label>
  <input type="text" id="name" name="name" />

  <label for="email">Email:</label>
```

```
<input type="email" id="email" name="email" />
```

```
<input type="submit" value="Submit" />  
</form>
```

Common Input Types:

text

email

password

checkbox

radio

submit

6. Tables in HTML

```
<table border="1">  
  <thead>  
    <tr>  
      <th>Name</th>  
      <th>Age</th>  
    </tr>  
  </thead>  
  <tbody>  
    <tr>  
      <td>Alice</td>  
      <td>24</td>  
    </tr>  
  </tbody>  
</table>
```

<table>: Table container.

<thead>, <tbody>, <tfoot>: Semantic grouping.

<tr>: Table row.

<th>: Header cell.

<td>: Data cell.

7. Semantic HTML Tags

<header>: Defines header for a document or section.

<nav>: Navigation links.

<main>: Main content.

<section>: Section of content.

<article>: Self-contained article.

<footer>: Footer of a document/section.

8. Media Elements

Audio

```
<audio controls>
  <source src="sound.mp3" type="audio/mpeg" />
  Your browser does not support audio.
</audio>
```

Video

```
<video width="320" height="240" controls>
  <source src="movie.mp4" type="video/mp4" />
  Your browser does not support video.
</video>
```

9. Meta Tags (SEO & Display)

```
<meta charset="UTF-8" />
<meta name="viewport" content="width=device-width, initial-scale=1.0" />
<meta name="description" content="Free HTML learning notes" />
```

Helps with responsive design and search engine optimization.